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Smart Donation Connect: An AI Based Platform for Reducing Waste and Connecting Donors with People in Need

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Abstract:

Millions of people in India are struggling to meet their daily needs even though there is plenty of surplus food, clothes, electronic devices and money available in the country. Every day, hotels, restaurants, event halls, businesses and households generate surplus food and other reusable items that are wasted. Meanwhile, orphanages, old-age homes, NGOs, poor families, and people living in remote villages are in dire need of resources. Although many wealthy individuals, business owners, and socially responsible citizens are willing to donate to charitable causes, they cannot find reliable beneficiaries since the needs of the underprivileged are hidden.

Similarly,many middle-class families have clothes and household items that are rarely used but can dramatically improve the lives of others.

This project proposes Smart Donation Connect – an AI-enabled web application that connects verified donors with verified beneficiaries nationwide. The platform allows individuals, hotels,restaurants, businesses, and organizations to donate surplus food, clothing, electronic items, and monetary assistance, while orphanages, old-age homes, NGOs, educational institutions, and economically disadvantaged families can register their requirements through a secure verification process.

The proposed system features an AI-based food demand prediction model that analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer behavior to estimate daily customer arrivals and surplus food quantities. This allows restaurants and hotels to optimize food preparation and reduce food wastage while distributing safe surplus food to nearby beneficiaries before it is discarded. The platform also features an AI-powered recommendation system that smartly matches donors with suitable recipients based on their location, urgency, donation type, and beneficiary requirements. Real-time notifications, transparent donation tracking, and verified user profiles foster trust among users. By connecting donors with people whose needs are often unseen, the proposed solution reduces waste, increases charitable engagement, and promotes sustainable development across the country.

Keywords: Artificial Intelligence; Web Development; Food Waste Prediction; Smart Donation Platform; Donation Matching; Resource Sharing; NGOs; Social Welfare; Real-Time Notifications; Sustainable Development.

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